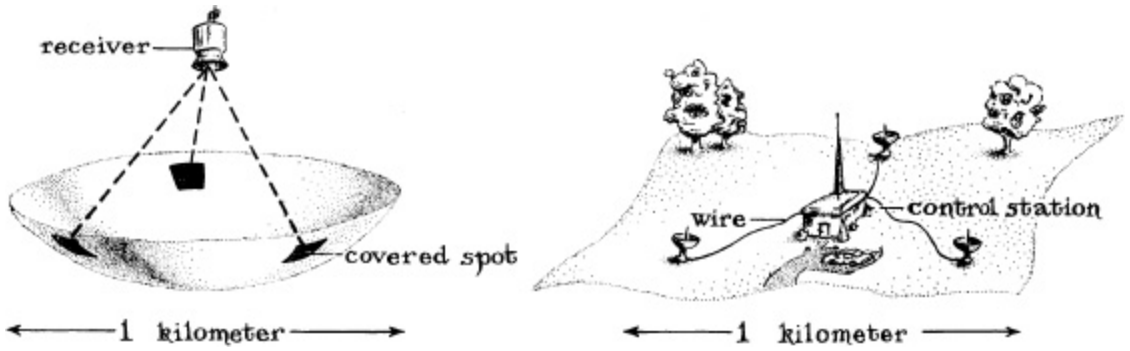
keeping only a few. In both cases, there is a modest loss of resolution, but a large loss of sensitivity. What the experimental physicists wanted most was an improved resolution (they wanted to find out where the radio waves were coming from and what the shapes of the radio sources were), not an improved sensitivity (not an ability to see more, dimmer radio sources-at least not for now). Therefore, they needed only a spotty bowl, not a fully covered bowl.



9.2 The principle of a radio interferometer. *Left:* In order to achieve good angular resolution, one would like to have a huge, say, l-kilometer, telescope. However, it would be sufficient if only a few spots (solid) on the radio-wave-reflecting bowl are actually covered with metal and reflect. *Right:* It is not necessary for the radio waves reflected from those spots to be focused to an antenna and radio receiver at the huge bowl's center. Rather, each spot can focus its waves to its own antenna and receiver, and the resulting radio signals can then be carried by wire from all the receivers to a central receiving station, where they are combined in the same manner as they would have been at the huge telescope's receiver. The result is a network of small radio telescopes with linked and combined outputs, a *radio interferometer.*

A practical way to make such a spotty bowl was by constructing a network of small radio telescopes connected by wires to a central radio receiving station (right half of Figure 9.2). Each small telescope was like a spot of metal on the big bowl, the wires carrying each small telescope's radio signal were like radio beams reflected from the big bowl's spots, and the central receiving station which combines the signals from the wires was like the big bowl's antenna and receiver, which combine the beams from the bowl's spots. Such networks of small telescopes, the centerpieces of the experimental physicists' efforts, were called *radio interferometers,* because the principle behind their operation was *interferometry:* By "interfering" the outputs of the small telescopes with each other in a manner we shall meet in Box 10.3 of Chapter 10, the central receiving station constructs a radio map or picture of the sky.